

SITRAKA L. RAVELOJAONA

AI & Machine Learning Engineer

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Open to international relocation · Visa sponsorship required

PROFILE

AI & ML Engineer with 3+ years building, deploying, and monitoring production GenAI and ML systems. Specialized in RAG and agentic architectures, LLM observability, and end-to-end MLOps on GCP and AWS. Delivery across multi-tenant AI SaaS, multimodal RAG (100+ daily conversations), computer vision (500K+ images), document intelligence, and anomaly detection. Strong in Python, hybrid retrieval (BM25 + embeddings), and LLM evaluation (LLM-as-Judge).

CORE SKILLS

GenAI & RAG: LangChain, LangGraph, LlamaIndex, PydanticAI, Agentic AI, MCP, hybrid search (BM25 + embeddings), pgvector, Qdrant, Pinecone, Weaviate, ChromaDB

LLMs: Gemini, OpenAI, Claude, Llama 3, Groq, Ollama; multi-model routing, prompt engineering, LLM-as-Judge, fine-tuning (LoRA/QLoRA/PEFT)

Machine Learning: XGBoost, scikit-learn, feature engineering, imbalanced data (SMOTE), cross-validation, hyperparameter optimization, Precision/Recall/F1/AUC-ROC

Deep Learning: PyTorch, TensorFlow, Transformers/Hugging Face, YOLO, ResNet, transfer learning, data augmentation

MLOps & Cloud: GCP Vertex AI, AWS SageMaker, Docker, MLflow, Weights & Biases, Langfuse, Arize Phoenix, CI/CD, Git, n8n

Data & ETL: Python (pandas, NumPy), SQL/PostgreSQL, Apache Spark, Apache Airflow, OCR (Doctr), Power BI

Languages: Python (FastAPI), SQL, JavaScript (React, Next.js)

PROFESSIONAL EXPERIENCE

AI & ML Engineer — RAG Architecture, Agents & Refactoring

Apr 2026 – Present

Matsiya (Superfastttt Platform) — Biarritz, France (Remote)

- Lead the audit and refactoring of 5 production RAG services (~10,700 LOC) on a multi-tenant AI SaaS platform (PydanticAI, LlamaIndex, pgvector, Qdrant, FastAPI, Next.js).
- Benchmarked agentic frameworks (PydanticAI, LangGraph, CrewAI, AutoGen) on multi-agent orchestration, MCP connector discovery, state management, and license compliance to steer platform direction.
- Designed KnowledgeSource v2: catalogue ingestion (JSON/PDF/MD) with Qdrant Cloud hybrid search (dense + BM25 sparse) and fail-closed cross-tenant isolation against data leakage.
- Built agent-discovery capability enabling agents to autonomously explore available tools, workflows, and MCP connectors.

AI & ML Engineer — GenAI, RAG, Agents & Computer Vision

Mar 2024 – Apr 2026

Malitix — Antananarivo, Madagascar

- Shipped an agentic multimodal RAG sales assistant (WhatsApp/Messenger) handling ~100 conversations/day across 800+ products; hybrid BM25 + multimodal embeddings on Qdrant reached 80% relevance, MVP delivered in 2 weeks.
- Automated a document-intelligence pipeline (~100 docs/day) via OCR (Doctr) + LLM, reaching 95% accuracy with per-field confidence scoring and LLM-as-Judge validation, replacing a manual process.
- Built computer-vision models: stamp/signature detection (YOLO, transfer learning, dataset augmented 100K → 500K images) and age classification (ResNet) for retail targeting.
- Engineered an anomaly-detection pipeline processing ~200 bank statements/day (OCR + Gemini) on private GCP Dataflow, designed for sensitive-data environments.
- Owned MLOps: LLM tracing (Langfuse), retriever monitoring (Arize Phoenix), and multi-agent orchestration (n8n).

Data Scientist

Jul 2023 – Feb 2024

Relia Consulting — Antananarivo, Madagascar

- Built an accounting chatbot (Rasa + TensorFlow): data preparation, intent classification, and end-to-end testing.
- Developed a stock-prediction system (linear regression, Adam optimizer) with experiment tracking in Weights & Biases.

Freelance AI Engineer & Data Analyst

Sep 2021 – Present

Multiple Clients — Remote

- Built an end-to-end RAG chatbot (Pinecone + Groq/Llama 3.3 70B + Botpress + Zapier) with automated lead capture, hot/warm/cold qualification scoring, and CRM integration (open source).
- Delivered healthcare analytics (MIS'AINA): strategic KPIs, Power BI dashboard, and a deterministic optimization algorithm for health-insurance brokerage.
- Optimized PostgreSQL for a government project (Mauritius), including social-benefits calculation modules (CNAPS).

SELECTED PROJECTS & R&D

- **Credit Scoring:** XGBoost anomaly detection on heavily imbalanced financial data; feature engineering, class rebalancing, FastAPI deployment.
- **Clinical Scribe:** fine-tuned Llama 3 8B (LoRA/QLoRA) for SOAP-format clinical notes with PII anonymization (Airflow, Transformers).
- **Cloud ML Pipelines:** automated training pipelines on GCP Vertex AI and AWS SageMaker.
- **Local LLM Benchmarking:** Ollama + Open WebUI + Docker for model benchmarking (Llama, Mistral) and tokens/sec optimization.

EDUCATION

Computer Science Engineering Degree

2018 – 2023

École Nationale d'Informatique, Université de Fianarantsoa · GPA 3.2/4.0

Specializations: Machine Learning, Deep Learning, Statistical Learning, Data Modeling.

CERTIFICATIONS & LANGUAGES

- Columbia University — Data Science & AI Specialization
- Finalist — Datatour 2025 Data Competition
- Languages: French (native), English (professional working)